

Date: Apr 25, 2026
To: "Eugeni Vidal-Tortosa" eugenividal@hotmail.com;eugeni.vidal@uab.cat
From: "Journal of Transport Geography" noreply_EMsupport@elsevier.com
Subject: Decision on submission to Journal of Transport Geography

Manuscript Number: **JTRG-D-26-00424**

Quantifying the accuracy of OpenStreetMap for longitudinal cycling-infrastructure change: A reproducible validation framework using Google Street View

Dear Dr. Vidal-Tortosa,

Thank you for submitting your manuscript to Journal of Transport Geography.

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following **major revisions**. I invite you to resubmit your manuscript after addressing the comments below. Please **resubmit** your revised manuscript **by Jun 24, 2026**.

To view your reviewer feedback, please log in as an author at <https://www.editorialmanager.com/jtrg/> and navigate to your manuscript in the "Submissions Needing Revision" folder under the Author Main Menu.

When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully. Please note that your revised submission may need to be re-reviewed.

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Journal of Transport Geography values your contribution and I look forward to receiving your revised manuscript.

Kind regards,
 Mengqiu Cao
 Associate Editor

Journal of Transport Geography

Reviewers' comments:

Reviewer #1:

The article lacks a clearly defined research objective, and it is more methodological in nature, which, of course, does not detract from its **substantive and scientific value**. The article analyses and validates data on the cycle route network from OSM and Google Street View. This is a **new and rather innovative approach**, which requires further clarification regarding:

Did the authors attempt to compare information on the **length of the cycle route network extracted from OSM with other data, e.g. from the city?**

Did the authors attempt to examine entire road sections using machine learning algorithms with the aid of GSV, or did they consider doing so? Did the authors use tools to extract data from OSM, and how did they select the road networks for each year from 2015 to 2023? Taking into account the network year by year and the changes could eliminate certain errors or even make them easier to spot.

The network maps show that in 2015, the layout was more skeletal and oriented in a band-like pattern towards the sea, whilst over the following 8 years, major road networks were built across it. But the sections that have been removed stand out. Isn't it perhaps the case that they were simply misclassified or were on a cycle lane (a lane on the carriageway), making them easy to remove from the data?

Finally, a very strong conclusion was drawn that the infrastructure in OSM, or the network itself, does a good job of incorporating new sections but struggles with those that have been removed. This is a very important conclusion for everyone who uses OSM data, not just linear data.

Reviewer #2:

This study aims to evaluate the accuracy of Open street Map in tracking cycling infrastructure change over time by employing a manual, Google Street View verification method in Barcelona using data from 2015 and 2023. **The aim of the study are pertinent to the field of cycling research and the methodological approach is sound**. My comments are fairly minor as I believe the manuscript to already be in a decent place.

Can the author **add to table 1** (or anywhere else in the manuscript) **the number of distinct cycling segments** (the total sample from which the GSV sample was derived)? I am perplexed as to why did the authors reviewed such a small number of cycling segments (n = 86) what percentage of all eligible segments does that represent? While it is true that manual GSV verifications of built environment feat 4 take time (I have conducted them many times for pedestrian and cycling infrastructures on hundreds of street segments each time), it is not a **justification to select such a small sample**. This represent my primary grievance with the paper as it is. I think **the authors should expand their sampling further** to cover as much as possible of the existing cycling network if they wish to be able to provide more generalizable comments on the precision of OSM for assessing changes in cycling infrastructure.

As a minor comment, there are some adjective choices in text that seem a bit odd or superfluous (could be from the use of AI) and it would

probably be best if they were cleaned out (e.g., "adjudicated baseline", "authoritative datasets")

Reviewer #3:

This study aims to assess the temporal accuracy of the OpenStreetMap (OSM) data by using historical Google Street View (GSV). I thoroughly enjoyed reading this manuscript; it is well structured, clearly written, and supported by a solid methodological approach. I support the publication of this manuscript and recommend that it undergo only very minor revisions before being reconsidered for publication, for the following reasons:

- The objectives are clearly articulated and address a significant methodological gap in the literature on longitudinal studies of cycling networks.
- The methodology is particularly rigorous, relying on a careful and innovative combined use of OpenStreetMap (OSM) and Google Street View (GSV) data.
- The results are robust and convincing, clearly presented, and supported by extensive and well-documented supplementary materials.

Minor revisions.

1. Title of Table 4 is missing.
2. Table 4 and Figure 5 present the same results. Including both is unnecessary and redundant. I suggest retaining only the table.
3. Table 3 reports 86 observations (Add:33; Remove: 5; NonCycl: 48), whereas Table 4 reports 76 observations (Add:33; Remove: 5; Pooled: 38). Is this difference expected? Please clarify.
4. The study is based on 86 Google Street View (GSV) locations assessed at two time points. This limitation should be discussed in Section 5 entitled Generalisability and Limitations.
5. It would be useful to discuss whether this methodological approach would remain feasible if applied to a much larger number of locations (e.g., 1,000 or 10,000 sites). The limitations related to Google Street View licensing terms are not discussed, including the prohibition of web scraping and the potential need to rely on the Google Street View API, along with possible cost implications.

Reviewer #4:

This paper addresses an important question: the reliability of OSM for detecting longitudinal cycling infrastructure change. The integration of OSM snapshot differencing with GSV-based validation is a clear strength, and the finding that additions are more reliable than removals is relevant. However, in its current form, the manuscript reads more like a technical report than a fully developed journal article, and several aspects of the validation design and interpretation require further clarification. In particular, the study raises a classic "proxy vs. proxy" issue in urban informatics, where one imperfect data source is used to validate another. While GSV provides a practical reference, it is itself a temporally and spatially uneven proxy. As such, it remains unclear to what extent the reported validation results reflect true agreement with ground truth, as opposed to inconsistencies between two imperfect datasets. This limitation should be more explicitly acknowledged and addressed in the manuscript.

Here are my detailed comments:

1. In Section 2.3.2 and Section 5, the manuscript relies on GSV imagery as a validation reference but does not sufficiently address its limitations. GSV coverage is uneven across space and time, and the use of fallback years (e.g., 2014 and 2022) introduces temporal mismatch with the OSM snapshots. In addition, GSV is implicitly treated as ground truth, although visibility and interpretability of cycling infrastructure vary by type. The authors should clarify how these issues may bias the reported precision and recall estimates and whether the set of usable validation points is systematically skewed. The authors should also look at few papers available that have discussed the coverage and biases of the GSV image.
2. In Section 2.3.1, the authors use a specific set of OSM tags to define cycling infrastructure, focusing on dedicated cycleways and certain lane/track attributes. This may exclude other forms of infrastructure (e.g., shared lanes) and could affect completeness. The authors should clarify how this definition influences the results and discusses implications for applying the framework in other contexts.
3. In Section 2.3.1 and Section 2.3.2, false negatives are identified using a NONCYC control pool that is explicitly filtered to exclude areas near cycling infrastructure. This design likely reduces the probability of detecting missed changes and may inflate recall estimates. The very high reported recall values (0.96-1.00) should therefore be interpreted with caution. The authors should clarify the assumptions underlying this design and discuss how it may affect the validity of recall as a performance metric.
4. The workflow depends on several parameter choices, including buffer distances, proximity thresholds, and minimum segment lengths (Section 2.3.1). While these are justified conceptually, the manuscript does not assess how sensitive the results are to these values. Given the claim of a transferable framework, the authors should either provide a sensitivity analysis or further justify the robustness of these parameter choices.
5. In Section 2.3.2 and also Table 3, although the sampling strategy is described as stratified, the actual implementation involves fixed numbers of samples per tract and is constrained by GSV availability. This leads to imbalances across classes (e.g., very few removals) and potential selection bias due to missing imagery (only 82% usable points). The authors should clarify whether the resulting validation set can be considered representative of the full set of detected changes.
6. The conclusions regarding removal detection are based on only five usable cases (Table 4), which is a very small sample. While this limitation is acknowledged, the low precision of removals is emphasized throughout the manuscript, including in the Abstract and Discussion. Given the wide uncertainty associated with such a small sample, the authors should moderate these claims or provide a stronger justification for their interpretation.
7. The manuscript emphasizes a "reproducible" and "transferable" framework; however, the validation step relies on manual coding of GSV imagery and depends on external data availability. As such, reproducibility is only partial, and transferability may vary across cities with different OSM data quality and GSV coverage. The authors should refine these claims accordingly.
8. Minor issues.
 - * There are minor inconsistencies in terminology (e.g., cycling infrastructure vs. cycleway) and some formatting/spacing issues throughout the manuscript that should be corrected.
 - * In addition, the structure of the Methods section could be improved for clarity. First, although the workflow is described as a five-step process, these steps are split across two subsections (Sections 2.3.1 and 2.3.2), which makes the overall logic somewhat difficult to follow. Second, the relationship between the two subsections (OSM differencing vs. GSV validation) is not clearly introduced at the beginning of the Methods section. Third, the ordering of steps across subsections requires the reader to move back and forth, which reduces readability.

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